In this lesson, you can find the full source code for the project used within the course. Feel free to use this lesson as needed – whether to help you debug your code, use it as a reference later, or expand the project to practice your skills!

app.py

Found in folder: **/**

This code defines a Flask web application that allows users to upload images, resize and rotate them, and apply a watermark. The application handles file uploads, validates the files, and processes the images according to user-specified parameters. The processed images are then saved and made available for download.

```
import os

from flask import Flask, render_template, request, send_from_directory, url_for, jsonify

from werkzeug.utils import secure_filename

from PIL import Image, ImageDraw, ImageFont

app = Flask(__name__)

app.config['UPLOAD_FOLDER'] = 'uploads'

app.config['MAX_CONTENT_LENGTH'] = 16 * 1024 * 1024  # 16MB max upload

ALLOWED_EXTENSIONS = {'png', 'jpg', 'jpeg'}

class WatermarkConfig:

  def __init__(self, text="ZenvaResizer", font_size=36,

         color=(255, 255, 255), opacity=128, position="center",

         rotation=0, padding=20):

    self.text = text

    self.font_size = font_size

    self.color = color

    self.opacity = opacity

    self.position = position

    self.rotation = rotation
```

```python
        self.padding = padding
def apply_watermark(image, config):
    """Apply watermark to the image"""
    # Create a transparent layer for the watermark
    watermark = Image.new('RGBA', image.size, (0, 0, 0, 0))
    draw = ImageDraw.Draw(watermark)

    # Load default font
    try:
        font = ImageFont.truetype("arial.ttf", config.font_size)
    except:
        font = ImageFont.load_default()

    # Calculate text size
    text_bbox = draw.textbbox((0, 0), config.text, font=font)
    text_width = text_bbox[2] - text_bbox[0]
    text_height = text_bbox[3] - text_bbox[1]

    # Calculate position based on config
    if config.position == "center":
        x = (image.size[0] - text_width) // 2
        y = (image.size[1] - text_height) // 2
    elif config.position == "top-left":
        x = config.padding
        y = config.padding
    elif config.position == "top-right":
```

```
    x = image.size[0] - text_width - config.padding
    y = config.padding
elif config.position == "bottom-left":
    x = config.padding
    y = image.size[1] - text_height - config.padding
elif config.position == "bottom-right":
    x = image.size[0] - text_width - config.padding
    y = image.size[1] - text_height - config.padding

# Draw text
draw.text((x, y), config.text, font=font, fill=(*config.color, config.opacity))

# Rotate watermark if needed
if config.rotation != 0:
    watermark = watermark.rotate(config.rotation, expand=True)
    # Recenter the rotated watermark
    x = (image.size[0] - watermark.size[0]) // 2
    y = (image.size[1] - watermark.size[1]) // 2
    watermark = watermark.crop((0, 0, image.size[0], image.size[1]))

# Convert image to RGBA if it isn't already
if image.mode != 'RGBA':
    image = image.convert('RGBA')

# Composite watermark onto image
return Image.alpha_composite(image, watermark)
```

```python
# Ensure upload directory exists
os.makedirs(app.config['UPLOAD_FOLDER'], exist_ok=True)
def allowed_file(filename):
    return '.' in filename and filename.rsplit('.', 1)[1].lower() in ALLOWED_EXTENSIONS
@app.route('/')
def index():
    return render_template('index.html')
@app.route('/upload', methods=['POST'])
def upload():
    if 'images[]' not in request.files:
        return jsonify({'error': 'No files uploaded'}), 400

    files = request.files.getlist('images[]')
    if len(files) > 5:
        return jsonify({'error': 'Maximum 5 files allowed'}), 400
    width = int(request.form.get('width', 800))
    height = int(request.form.get('height', 600))
    quality = int(request.form.get('quality', 85))
    rotation = int(request.form.get('rotation', 0))
    # Get watermark parameters
    watermark_text = request.form.get('watermark_text', 'ZenvaResizer')
    font_size = int(request.form.get('font_size', 36))
    color = tuple(map(int, request.form.get('color', '255,255,255').split(',')))
    opacity = int(request.form.get('opacity', 128))
    position = request.form.get('position', 'center')
    watermark_rotation = int(request.form.get('watermark_rotation', 0))
```

```
padding = int(request.form.get('padding', 20))
# Create watermark configuration
watermark_config = WatermarkConfig(
   text=watermark_text,
   font_size=font_size,
   color=color,
   opacity=opacity,
   position=position,
   rotation=watermark_rotation,
   padding=padding
)
processed_files = []
for file in files:
   if file and allowed_file(file.filename):
      filename = secure_filename(file.filename)
      input_path = os.path.join(app.config['UPLOAD_FOLDER'], filename)
      output_filename = f'resized_{filename}'
      output_path = os.path.join(app.config['UPLOAD_FOLDER'], output_filename)

      file.save(input_path)

      with Image.open(input_path) as img:
         # Apply rotation if specified
         if rotation != 0:
            img = img.rotate(rotation, expand=True, resample=Image.Resampling.BICUBIC)
         # Resize after rotation to maintain aspect ratio
```

```
                img = img.resize((width, height), Image.Resampling.LANCZOS)

                # Apply watermark
                img = apply_watermark(img, watermark_config)

                # Save the processed image
                img.save(output_path, quality=quality)

            processed_files.append(output_filename)
            os.remove(input_path)  # Clean up original file
    if request.headers.get('X-Requested-With') == 'XMLHttpRequest':
        return jsonify({'success': True, 'files': processed_files})
    return render_template('results.html', files=processed_files)
@app.route('/uploads/<filename>')
def uploaded_file(filename):
    return send_from_directory(app.config['UPLOAD_FOLDER'], filename)
if __name__ == '__main__':
    app.run(debug=True)
```

index.html

Found in folder: **/templates**

This HTML code defines a web page for an image resizer application, including a form for users to upload images and specify resizing options, as well as a section to display the processed images. The code also includes JavaScript that handles form submission, sends a request to a server to process the images, and updates the page with the processed images. The page has a dark theme with neon-colored accents and includes a footer with links and copyright information.

```
<!DOCTYPE html>
```

```
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Image Resizer</title>
  <style>
    :root {
      --neon-pink: #ff2e88;
      --neon-blue: #00f3ff;
      --neon-purple: #9d00ff;
      --dark-bg: #0a0a0a;
    }

    body {
      font-family: 'Segoe UI', Arial, sans-serif;
      max-width: 800px;
      margin: 0 auto;
      padding: 20px;
      background-color: var(--dark-bg);
      color: #fff;
      line-height: 1.6;
      min-height: 100vh;
      display: flex;
      flex-direction: column;
    }
    header {
```

```css
  text-align: center;
  padding: 20px 0;
  margin-bottom: 40px;
  border-bottom: 1px solid rgba(255, 255, 255, 0.1);
}
.logo {
  font-size: 2.5em;
  font-weight: bold;
  color: var(--neon-blue);
  text-shadow: 0 0 10px var(--neon-blue),
         0 0 20px var(--neon-blue),
         0 0 30px var(--neon-blue);
  margin: 0;
  letter-spacing: 2px;
}
.tagline {
  color: var(--neon-pink);
  margin-top: 10px;
  font-size: 1.1em;
  opacity: 0.9;
}
footer {
  margin-top: auto;
  text-align: center;
  padding: 20px 0;
  border-top: 1px solid rgba(255, 255, 255, 0.1);
```

```css
    color: rgba(255, 255, 255, 0.6);
}
.footer-links {
    margin: 10px 0;
}
.footer-links a {
    color: var(--neon-blue);
    text-decoration: none;
    margin: 0 10px;
    transition: all 0.3s ease;
}
.footer-links a:hover {
    color: var(--neon-pink);
    text-shadow: 0 0 10px var(--neon-pink);
}
.form-group {
    margin-bottom: 25px;
    background: rgba(255, 255, 255, 0.05);
    padding: 20px;
    border-radius: 8px;
    border: 1px solid rgba(255, 255, 255, 0.1);
    transition: all 0.3s ease;
}
.form-group:hover {
    border-color: var(--neon-purple);
    box-shadow: 0 0 15px rgba(157, 0, 255, 0.3);
```

```
}
label {
  display: block;
  margin-bottom: 10px;
  color: var(--neon-pink);
  font-weight: 500;
}
input[type="number"],
input[type="text"],
select {
  width: 100%;
  padding: 8px 12px;
  background: rgba(255, 255, 255, 0.1);
  border: 1px solid var(--neon-blue);
  border-radius: 4px;
  color: #fff;
  font-size: 16px;
  margin-bottom: 10px;
}
input[type="number"]:focus,
input[type="text"]:focus,
select:focus {
  outline: none;
  box-shadow: 0 0 10px var(--neon-blue);
}
input[type="file"] {
```

```
  background: rgba(255, 255, 255, 0.1);
  padding: 10px;
  border-radius: 4px;
  border: 1px dashed var(--neon-blue);
  color: #fff;
  width: 100%;
}
input[type="file"]::-webkit-file-upload-button {
  background: var(--neon-purple);
  color: white;
  padding: 8px 16px;
  border: none;
  border-radius: 4px;
  cursor: pointer;
  margin-right: 10px;
}
input[type="color"] {
  width: 100%;
  height: 40px;
  padding: 2px;
  background: rgba(255, 255, 255, 0.1);
  border: 1px solid var(--neon-blue);
  border-radius: 4px;
  cursor: pointer;
}
input[type="range"] {
```

```css
    width: 100%;
    height: 8px;
    background: rgba(255, 255, 255, 0.1);
    border-radius: 4px;
    outline: none;
    -webkit-appearance: none;
}
input[type="range"]::-webkit-slider-thumb {
    -webkit-appearance: none;
    width: 20px;
    height: 20px;
    background: var(--neon-purple);
    border-radius: 50%;
    cursor: pointer;
    transition: all 0.3s ease;
}
input[type="range"]::-webkit-slider-thumb:hover {
    background: var(--neon-pink);
    box-shadow: 0 0 10px var(--neon-pink);
}
button {
    background: linear-gradient(45deg, var(--neon-purple), var(--neon-pink));
    color: white;
    padding: 12px 30px;
    border: none;
    border-radius: 25px;
```

```css
  cursor: pointer;
  font-size: 16px;
  font-weight: bold;
  width: 100%;
  text-transform: uppercase;
  letter-spacing: 1px;
  transition: all 0.3s ease;
  box-shadow: 0 0 15px rgba(157, 0, 255, 0.3);
}
button:hover {
  transform: translateY(-2px);
  box-shadow: 0 0 20px rgba(157, 0, 255, 0.5);
}
.container {
  max-width: 800px;
  margin: 0 auto;
  padding: 20px;
  flex: 1;
}
.watermark-controls {
  background: rgba(255, 255, 255, 0.05);
  padding: 20px;
  border-radius: 8px;
  border: 1px solid rgba(255, 255, 255, 0.1);
  margin-bottom: 20px;
  transition: all 0.3s ease;
```

```css
}
.watermark-controls:hover {
  border-color: var(--neon-purple);
  box-shadow: 0 0 15px rgba(157, 0, 255, 0.3);
}
.watermark-controls h4 {
  color: var(--neon-blue);
  margin-bottom: 20px;
  text-shadow: 0 0 10px var(--neon-blue);
}
.preview-container {
  max-width: 300px;
  margin: 20px auto;
  position: relative;
}
.preview-image {
  max-width: 100%;
  border: 1px solid var(--neon-blue);
  border-radius: 4px;
  box-shadow: 0 0 10px rgba(0, 243, 255, 0.3);
}
.results {
  margin-top: 30px;
}
.results h3 {
  color: var(--neon-blue);
```

```css
    text-shadow: 0 0 10px var(--neon-blue);
    margin-bottom: 20px;
}
.download-btn {
    background: linear-gradient(45deg, var(--neon-purple), var(--neon-pink));
    color: white;
    padding: 8px 20px;
    border: none;
    border-radius: 20px;
    cursor: pointer;
    font-size: 14px;
    text-decoration: none;
    display: inline-block;
    margin-top: 10px;
    transition: all 0.3s ease;
}
.download-btn:hover {
    transform: translateY(-2px);
    box-shadow: 0 0 15px rgba(157, 0, 255, 0.5);
}
.opacity-value {
    color: var(--neon-pink);
    text-align: center;
    margin-top: 5px;
    font-size: 14px;
}
```

```html
    </style>
</head>
<body>
    <header>
        <h1 class="logo">ZenvaResizer</h1>
        <p class="tagline">Transform your images with neon precision</p>
    </header>
    <div class="container">
        <form id="uploadForm" enctype="multipart/form-data">
            <div class="form-group">
                <label for="images">Select Images (Max 5)</label>
                <input type="file" id="images" name="images[]" multiple accept="image/*" required>
            </div>
            <div class="form-group">
                <label for="width">Width (px)</label>
                <input type="number" id="width" name="width" value="800" required>
            </div>
            <div class="form-group">
                <label for="height">Height (px)</label>
                <input type="number" id="height" name="height" value="600" required>
            </div>
            <div class="form-group">
                <label for="quality">Quality (1-100)</label>
                <input type="number" id="quality" name="quality" value="85" min="1" max="100" required>
            </div>
```

```
      <div class="form-group">
        <label for="rotation">Rotation (degrees)</label>
        <input type="number" id="rotation" name="rotation" value="0" min="0" max="360">
      </div>
      <div class="watermark-controls">
        <h4>Watermark Settings</h4>

        <div class="form-group">
          <label for="watermark_text">Watermark Text</label>
          <input type="text" id="watermark_text" name="watermark_text" placeholder="Enter watermark text">
        </div>
        <div class="form-group">
          <label for="font_size">Font Size</label>
          <input type="number" id="font_size" name="font_size" value="36" min="8" max="200">
        </div>
        <div class="form-group">
          <label for="color">Color</label>
          <input type="color" id="color" name="color" value="#ffffff">
        </div>
        <div class="form-group">
          <label for="opacity">Opacity (0-255)</label>
          <input type="range" id="opacity" name="opacity" min="0" max="255" value="128">
          <div class="opacity-value" id="opacityValue">128</div>
```

```
        </div>
        <div class="form-group">
          <label for="position">Position</label>
          <select id="position" name="position">
            <option value="center">Center</option>
            <option value="top-left">Top Left</option>
            <option value="top-right">Top Right</option>
            <option value="bottom-left">Bottom Left</option>
            <option value="bottom-right">Bottom Right</option>
          </select>
        </div>
        <div class="form-group">
          <label for="watermark_rotation">Watermark Rotation (degrees)</label>
          <input type="number" id="watermark_rotation" name="watermark_rotation"
value="0" min="0" max="360">
        </div>
        <div class="form-group">
          <label for="padding">Padding (px)</label>
          <input type="number" id="padding" name="padding" value="20" min="0"
max="100">
        </div>
      </div>
      <button type="submit">Process Images</button>
    </form>
    <div id="results" class="results"></div>
  </div>
  <footer>
```

```
    <div class="footer-links">
      <a href="/">Home</a>
      <a href="https://github.com" target="_blank">GitHub</a>
      <a href="https://pillow.readthedocs.io" target="_blank">Pillow Docs</a>
    </div>
    <p>&copy; 2025 ZenvaResizer. All rights reserved.</p>
  </footer>
  <script>
    document.getElementById('opacity').addEventListener('input', function() {
      document.getElementById('opacityValue').textContent = this.value;
    });
    document.getElementById('uploadForm').addEventListener('submit', function(e) {
      e.preventDefault();

      const formData = new FormData(this);
      const color = document.getElementById('color').value;
      formData.set('color', `${parseInt(color.substr(1,2), 16)},${parseInt(color.substr(3,2),
16)},${parseInt(color.substr(5,2), 16)}` );
      fetch('/upload', {
        method: 'POST',
        body: formData,
        headers: {
          'X-Requested-With': 'XMLHttpRequest'
        }
      })
      .then(response => response.json())
```

```
        .then(data => {
          if (data.success) {
            const resultsDiv = document.getElementById('results');
            resultsDiv.innerHTML = '<h3>Processed Images:</h3>';
            data.files.forEach(file => {
              resultsDiv.innerHTML += `
                <div class="form-group">
                  <img src="/uploads/${file}" class="preview-image" alt="Processed image">
                  <a href="/uploads/${file}" download class="download-btn">Download</a>
                </div>
              `;
            });
          } else {
            alert('Error: ' + data.error);
          }
        })
        .catch(error => {
          console.error('Error:', error);
          alert('An error occurred while processing the images.');
        });
      });
  </script>
</body>
</html>
```

results.html

Found in folder: **/templates**

This HTML code defines the structure and styling of a web page for displaying resized images. It includes a header with a logo and tagline, a container for displaying the resized images with download links, and a footer with navigation links and copyright information. The page uses CSS styles to create a neon-themed design with custom colors and effects.

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Resized Images - ZenvaResizer</title>

  <style>

    :root {

      --neon-pink: #ff2e88;

      --neon-blue: #00f3ff;

      --neon-purple: #9d00ff;

      --dark-bg: #0a0a0a;

    }

    body {

      font-family: 'Segoe UI', Arial, sans-serif;

      max-width: 800px;

      margin: 0 auto;

      padding: 20px;

      background-color: var(--dark-bg);

      color: #fff;

      line-height: 1.6;

      min-height: 100vh;

      display: flex;
```

```
  flex-direction: column;
}
header {
  text-align: center;
  padding: 20px 0;
  margin-bottom: 40px;
  border-bottom: 1px solid rgba(255, 255, 255, 0.1);
}
.logo {
  font-size: 2.5em;
  font-weight: bold;
  color: var(--neon-blue);
  text-shadow: 0 0 10px var(--neon-blue),
         0 0 20px var(--neon-blue),
         0 0 30px var(--neon-blue);
  margin: 0;
  letter-spacing: 2px;
}
.tagline {
  color: var(--neon-pink);
  margin-top: 10px;
  font-size: 1.1em;
  opacity: 0.9;
}
footer {
  margin-top: auto;
```

```css
  text-align: center;
  padding: 20px 0;
  border-top: 1px solid rgba(255, 255, 255, 0.1);
  color: rgba(255, 255, 255, 0.6);
}
.footer-links {
  margin: 10px 0;
}
.footer-links a {
  color: var(--neon-blue);
  text-decoration: none;
  margin: 0 10px;
  transition: all 0.3s ease;
}
.footer-links a:hover {
  color: var(--neon-pink);
  text-shadow: 0 0 10px var(--neon-pink);
}
.image-container {
  margin: 30px 0;
  padding: 20px;
  background: rgba(255, 255, 255, 0.05);
  border: 1px solid rgba(255, 255, 255, 0.1);
  border-radius: 8px;
  transition: all 0.3s ease;
}
```

```
.image-container:hover {
  border-color: var(--neon-purple);
  box-shadow: 0 0 20px rgba(157, 0, 255, 0.3);
  transform: translateY(-2px);
}
.image-container img {
  max-width: 100%;
  height: auto;
  border-radius: 4px;
  box-shadow: 0 0 15px rgba(0, 243, 255, 0.2);
}
.download-link {
  display: inline-block;
  background: linear-gradient(45deg, var(--neon-purple), var(--neon-pink));
  color: white;
  padding: 12px 30px;
  text-decoration: none;
  border-radius: 25px;
  margin-top: 15px;
  font-weight: bold;
  text-transform: uppercase;
  letter-spacing: 1px;
  transition: all 0.3s ease;
  box-shadow: 0 0 15px rgba(157, 0, 255, 0.3);
}
.download-link:hover {
```

```css
  transform: translateY(-2px);
  box-shadow: 0 0 20px rgba(157, 0, 255, 0.5);
}
.back-link {
  display: inline-block;
  margin-bottom: 30px;
  color: var(--neon-blue);
  text-decoration: none;
  font-weight: 500;
  transition: all 0.3s ease;
  padding: 8px 16px;
  border-radius: 4px;
  background: rgba(0, 243, 255, 0.1);
}
.back-link:hover {
  color: var(--neon-pink);
  background: rgba(255, 46, 136, 0.1);
  text-shadow: 0 0 10px var(--neon-pink);
}
.container {
  max-width: 600px;
  margin: 0 auto;
  padding: 20px;
  flex: 1;
}
.page-title {
```

```
        text-align: center;
        color: var(--neon-blue);
        text-shadow: 0 0 10px var(--neon-blue),
                     0 0 20px var(--neon-blue),
                     0 0 30px var(--neon-blue);
        margin-bottom: 40px;
      }
    </style>
</head>
<body>
    <header>
      <h1 class="logo">ZenvaResizer</h1>
      <p class="tagline">Transform your images with neon precision</p>
    </header>
    <div class="container">
      <a href="/" class="back-link">← Back to Upload</a>
      <h2 class="page-title">Resized Images</h2>

      {% for file in files %}
      <div class="image-container">
        <img src="{{ url_for('uploaded_file', filename=file) }}" alt="Resized image">
        <br>
        <a href="{{ url_for('uploaded_file', filename=file) }}" download class="download-link">Download</a>
      </div>
      {% endfor %}
```

```
  </div>
  <footer>
    <div class="footer-links">
      <a href="/">Home</a>
      <a href="https://github.com" target="_blank">GitHub</a>
      <a href="https://pillow.readthedocs.io" target="_blank">Pillow Docs</a>
    </div>
    <p>&copy; 2024 ZenvaResizer. All rights reserved.</p>
  </footer>
</body>
</html>
```

requirements.txt

Found in folder: **/**

This text document lists the dependencies required for a project, specifying the exact version of each library. The dependencies include Flask, Pillow, Werkzeug, and requests. The versions of these libraries are pinned to ensure consistency and reproducibility.

```
Flask==2.0.1
Pillow==9.0.0
Werkzeug==2.0.1
requests==2.31.0
```

README.md

Found in folder: **/**

This is a Cursor generated README file written in Markdown. It provides information about the ZenvaResizer project, including its features, requirements, installation, usage, API endpoints, and license. The file serves as a guide for users and developers who want to use or contribute to the project.

```
# ZenvaResizer
```

A modern, neon-themed web application **for** batch image processing with a sleek user interface. Built with Flask and Pillow, ZenvaResizer allows you to resize, rotate, and optimize multiple images simultaneously.

Features

- 🖼️ **Batch Processing**: Upload and process up to 5 images at once

- 📐 **Custom Dimensions**: Set custom width and height **for** your images

- 🔄 **Rotation**: Rotate images by any angle (0-359 degrees)

- 🎯 **Quality Control**: Adjust output image quality (1-100)

- 🎨 **Modern UI**: Sleek neon-themed interface with smooth animations

- 🚀 **Fast Processing**: Efficient image processing using Pillow

- 📱 **Responsive Design**: Works on both desktop and mobile devices

Requirements

- Python 3.7+

- Flask

- Pillow (PIL)

- Werkzeug

Installation

1. Clone the repository:

```bash
git clone https://github.com/yourusername/zenvaresizer.git
cd zenvaresizer
```

2. Create and activate a virtual environment:

```bash
python -m venv venv
source venv/bin/activate  # On Windows: venv\Scripts\activate
```

```

3. Install dependencies:

```bash

pip install -r requirements.txt

```

Usage

1. Start the application:

```bash

python app.py

```

2. Open your web browser and navigate to `http://localhost:5000`

3. Use the web interface to:

  - Select up to 5 images

  - Set desired dimensions

  - Adjust quality settings

  - Apply rotation **if** needed

  - Process and download the results

API Endpoints

- `GET /`: Main application interface

- `POST /upload`: Process uploaded images

  - Parameters:

    - `images**[]**`: Image files **(**max 5**)**

    - `width`: Target width **in** pixels

    - `height`: Target height **in** pixels

    - `quality`: Output quality **(**1-100**)**

    - `rotation`: Rotation angle **in** degrees **(**0-359**)**
```

- ` GET /uploads/<filename>` : Access processed images

Supported Image Formats

- PNG

- JPEG/JPG

Contributing

1. Fork the repository

2. Create your feature branch (` git checkout -b feature/AmazingFeature`)

3. Commit your changes (` git commit -m 'Add some AmazingFeature'`)

4. Push to the branch (` git push origin feature/AmazingFeature`)

5. Open a Pull Request

License

This project is licensed under the MIT License - see the LICENSE file for details.

Acknowledgments

- [Flask](https://flask.palletsprojects.com/)

- [Pillow](https://python-pillow.org/)

- [Werkzeug](https://werkzeug.palletsprojects.com/)

Contact

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Project Link: [https://github.com/yourusername/zenvaresizer](https://github.com/youruser